

Anatomical landmarks

On the tooth surface

Landmarks mean Elevations, depressions and grooves are present on the surfaces of teeth, and different terminologies are used to describe these landmarks.

To study an individual tooth intelligently, we must be able to recognize all landmarks of importance by name. These include

- **Concave:** A curvature that leans inward
- **Convex:** A curvature that extends outward. Opposite of concave.

Anatomical Landmarks on the Crown Surface:

1. **Lobe:** Developmental lobe is the growth center in the development of the crown of the tooth.

- ❖ Each tooth begins to develop from four or more growth centers
- ❖ These centers are known as developmental lobes then unite together to form the crown of the tooth.
- ❖ Cusps and mamelons are representative of lobes.

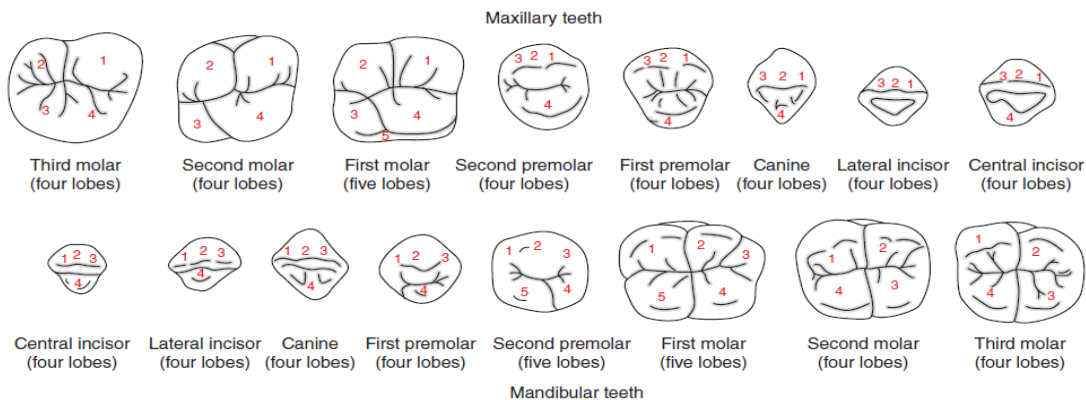


Figure 9.16 Developmental lobes.

2. **Mamelons:** are the protrusions seen in the incisal ridges of newly erupted permanent incisor teeth.

✚ They represent the three lobes from which the labial surface forms, and they are separated by developmental grooves.

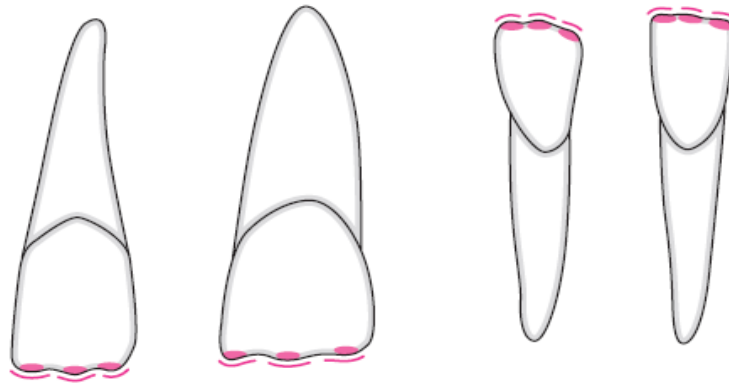


Figure 9.17 Mamelons in the anterior teeth.

3. **Cingulum** (Latin word for girdle)

- It is the lingual lobe of an anterior tooth.
- It makes up the bulk of the cervical third of the lingual surface (a prominence of enamel).
- Its convexity mesiodistally resembles a girdle encircling the lingual surface at the cervical third.

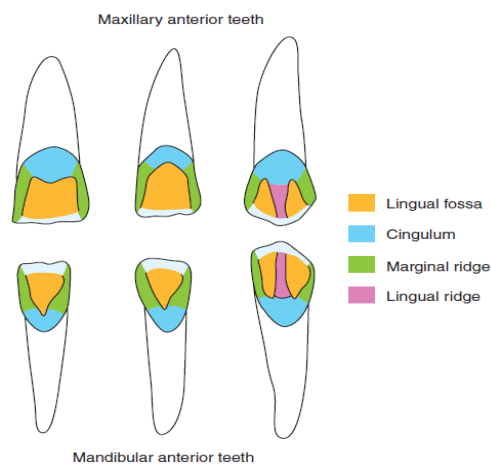
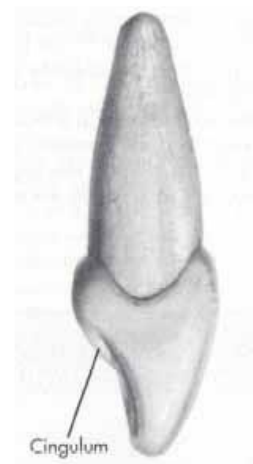


Figure 9.15 Lingual aspect of the anterior teeth.

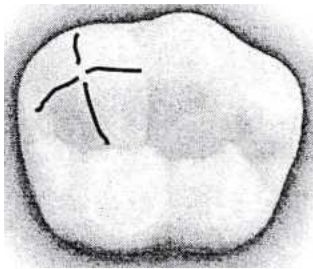


4. Cusp:

it is an elevation or mound on the crown portion of a tooth making up a divisional part of the occlusal surface.

These are conical or pyramidal projections on the crown portion of the tooth. Cusps are found on premolars & molars as well as canines.

It denotes the primary lobe during development. When the crown matured the cusp is formed of enamel, dentine, and pulp horn



5. Tubercle:

- It is a small, rounded prominence on the surface of a tooth resulting from the extra formation of enamel.
- deviations from the normal form. For example, the cusp of Carabelli is a tubercle.
- **The cusp of Carabelli** is a fifth cusp present on the palatal surface of the maxillary first molars.

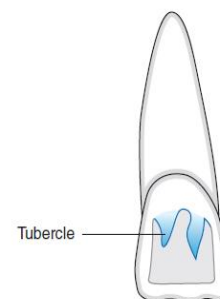


Figure 9.14 Permanent maxillary lateral incisor with tubercles.

6. Ridge:

Ridges are any linear, flat elevations on teeth, and they are named according to their location or form.

A) Marginal ridges: these are rounded borders of the enamel that form the mesial and distal margins of the occlusal surfaces of premolars and molars and the mesial and distal margins of the lingual surfaces of the incisors and canines.

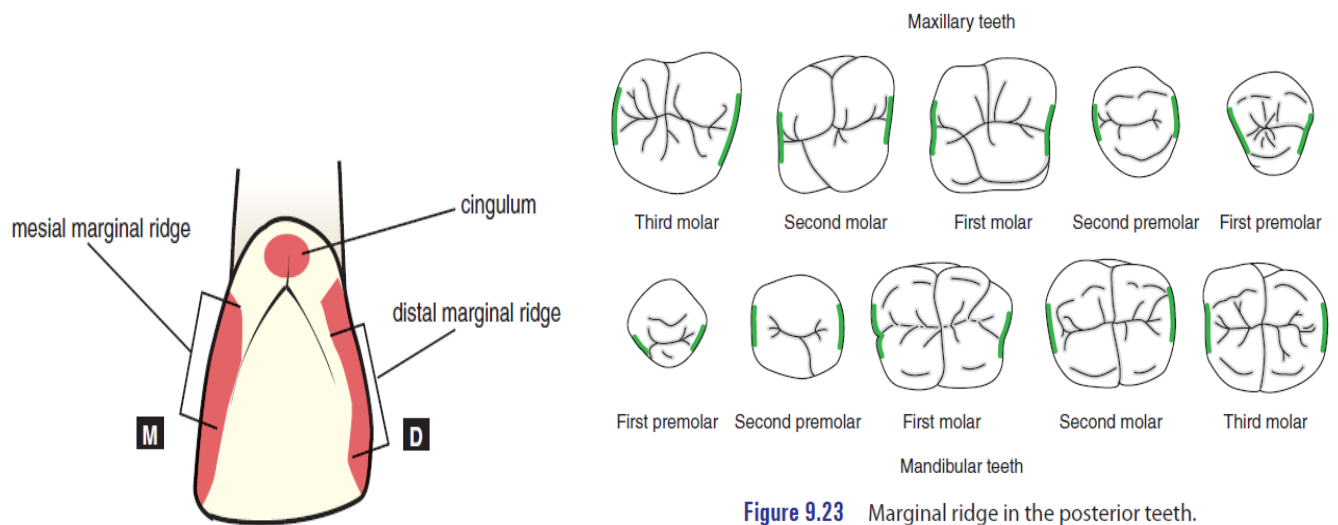
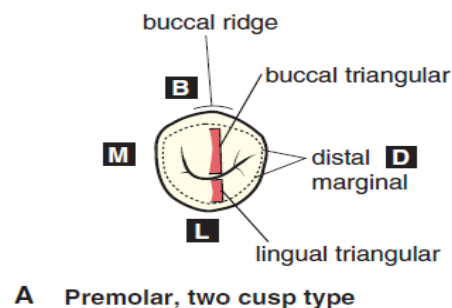
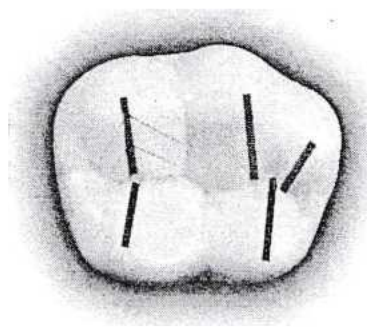


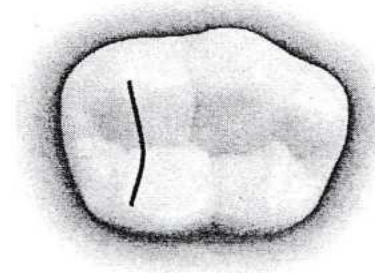
Figure 9.23 Marginal ridge in the posterior teeth.

B) Triangular ridges: these descend from the tips of the cusps of molars and premolars toward the central part of the occlusal surfaces. They are so named because the slopes of each side of the ridge are inclined to resemble two sides of a triangle. They are named after the cusps to which they belong, e.g., the triangular ridge of the buccal cusp of the maxillary first premolar.



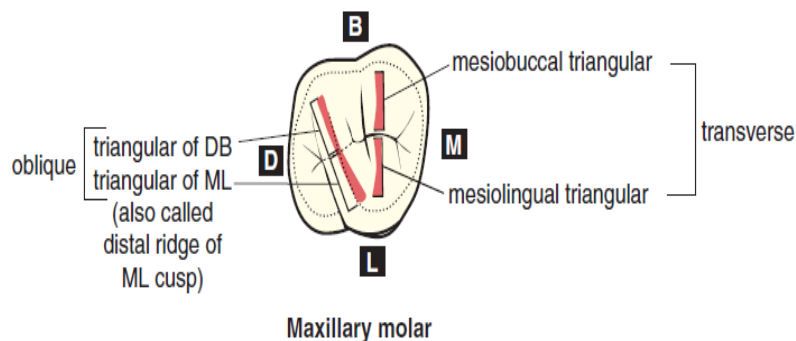
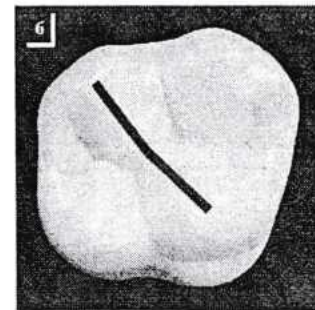
C) Transverse ridges:

it is the union of two triangular ridges crossing transversely the surface of a posterior tooth.



D) Oblique ridges:

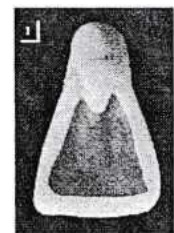
it is a ridge crossing obliquely the occlusal surfaces of maxillary molars. It is formed by the union of the triangular ridge of the distobuccal cusp and the distal cusp ridge of the mesiolingual cusp.



7. Sulcus: it is a long depression or valley in the surface of a tooth between ridges and cusps, the inclines of which meet at an angle. A sulcus has a developmental groove at the junction of its inclines.

8. Fossa: it is an irregular depression or concavity.

(a) Lingual fossa: it is located on the lingual surface of anterior teeth.



(b) Central fossa: it is located on the occlusal surface of the molars.

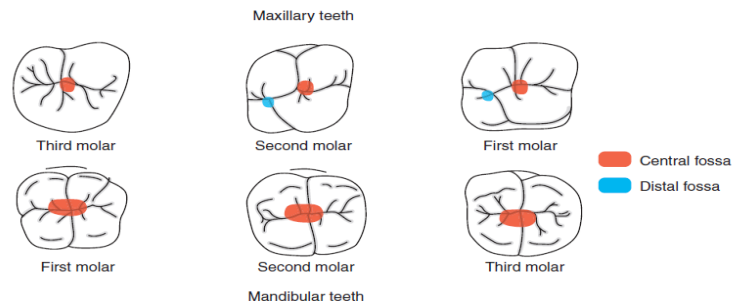


Figure 9.28 Central and distal fossa.

(C) Triangular fossa: it is located on the occlusal surfaces of molars and premolars, mesial or distal to marginal ridges.

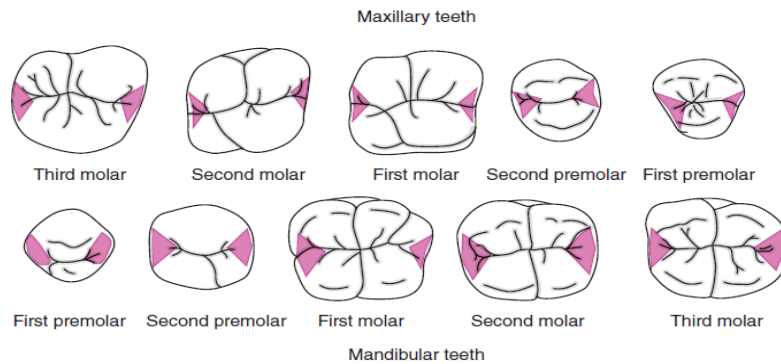
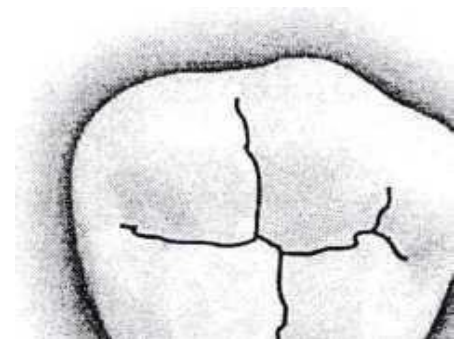
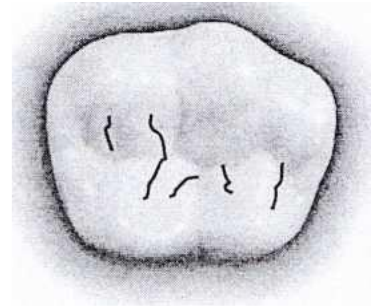


Figure 9.29 Mesial and distal triangular fossa.

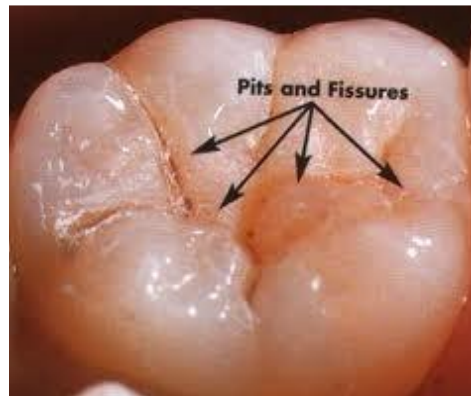
9. Developmental groove: it is a shallow groove or line between the primary parts of the crown or root.



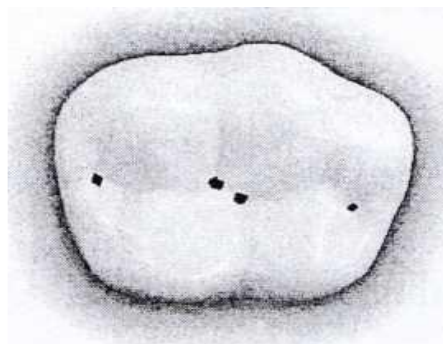
10. Supplemental groove: it is a less distinct, shallow linear depression on the surface of a tooth, but is supplemental to a developmental groove and does not mark the junction of primary parts.



11. Fissure: It is a narrow channel or cleft which is formed at the depth of a developmental groove and results from the incomplete union of the primary lobes. It is considered a fault or abnormality in enamel. Decay (caries) often begins in a deep fissure.

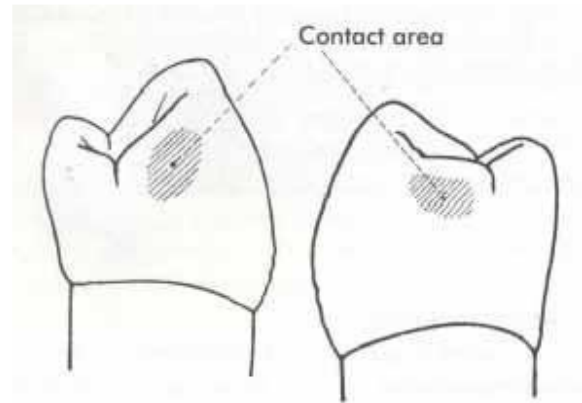
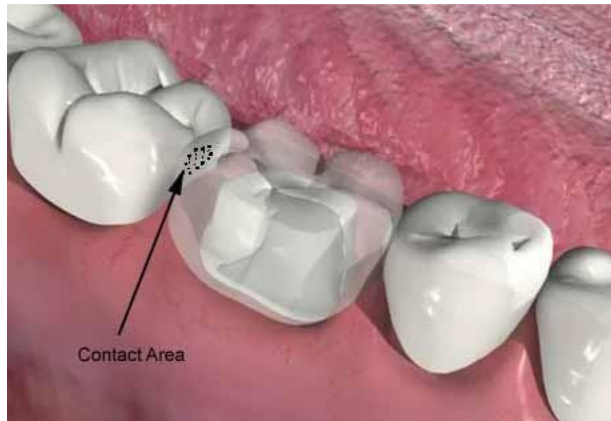


12. Pit: it is a small pinpoint depression located at the junction of developmental grooves or at terminals of those grooves, e.g., the central pit is a term used to describe a landmark in the central fossa of molars where developmental grooves join.



13. Contact area:

The area of a tooth that physically touches the abutment tooth; the contact areas occur in the proximal surfaces of teeth.



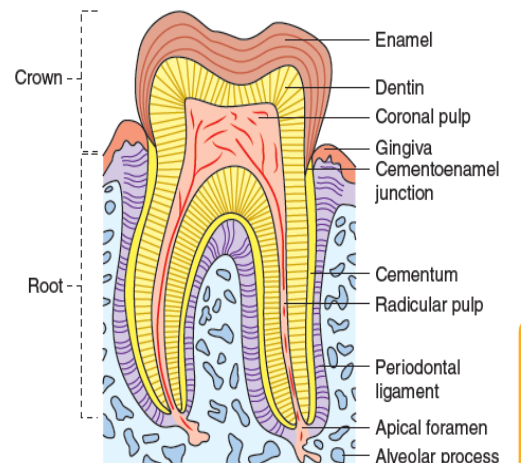
Anatomical Landmarks on the Root Surface

Apex

The terminal end of the root portion of the tooth is termed the *apex*.

Apical Foramen and Accessory Foramen

The apical foramen is the opening of the pulp canal at the apical end of the root through which blood and nerve supply passes into the pulp canal



Root Trunk

In multirooted teeth, the undivided cervical portion of the root is termed the *root trunk*.

Furcation

In multirooted teeth, the division of the root is termed *furcation*. It is called bifurcation if the root is divided into two parts and trifurcation if the root is divided into three parts.

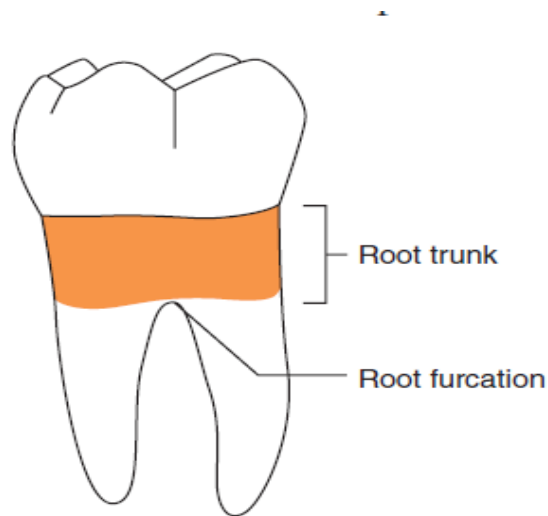


Figure 9.36 Root trunk and furcation of the mandibular first molar.